Ml-DL-Ops Assignment 3

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Objective

The objective of this assignment is to build an end-to-end machine learning pipeline including model training, evaluation, Hugging Face deployment, and Docker-based reproducible execution for review genre classification.

Model Selection

The DistilBERT (distilbert-base-cased) model was selected because:

- Lightweight and computationally efficient transformer
- Faster training than BERT
- Good performance on text classification tasks
- Suitable for deployment environments

DistilBERT uses a distilled transformer architecture that preserves contextual understanding with fewer parameters.

Training Summary

- Dataset built from multi-genre review data
- Tokenization using DistilBERT tokenizer (max length 512)
- Training using Hugging Face Trainer API
- 3 training epochs
- Batch size: 8 (train), 16 (eval)

Results

Overall Performance

Metric	Value
Accuracy	0.625
Loss	1.1115
Macro F1 Score	0.6286
Weighted F1 Score	0.6286

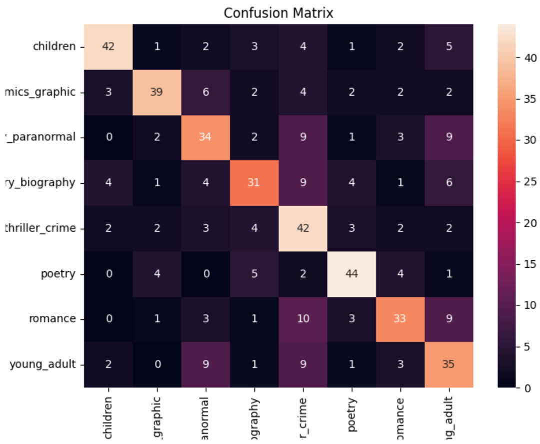
Class-wise Performance

Genre	Precision	Recall	F1
Children	0.79	0.70	0.74
Comics	0.78	0.65	0.71
Fantasy	0.56	0.57	0.56
History	0.63	0.52	0.57
Mystery	0.47	0.70	0.56
Poetry	0.75	0.73	0.74
Romance	0.66	0.55	0.60
Young Adult	0.51	0.58	0.54

Visualizations

Training and evaluation behavior is visualized using loss curves and confusion matrix.

Training and Validation Loss



Confusion Matrix

Docker Deployment

- Converted notebook to modular scripts
- Created Docker container with required dependencies
- Container loads model and performs evaluation automatically
- Ensures reproducible execution environment

Challenges

- Cleaning notebook and organizing scripts
- Managing Docker dependencies
- Handling large text sequences
- Ensuring consistent evaluation across environments

Links

- Hugging Face model : `https://huggingface.co/arpita2desh/distilbert-genres-mlops`
- Github : `https://github.com/arpitadeshmukh/MLOps-Arpita_Abhijit_Deshmukh-B23CM1007/tree/Assignment3`